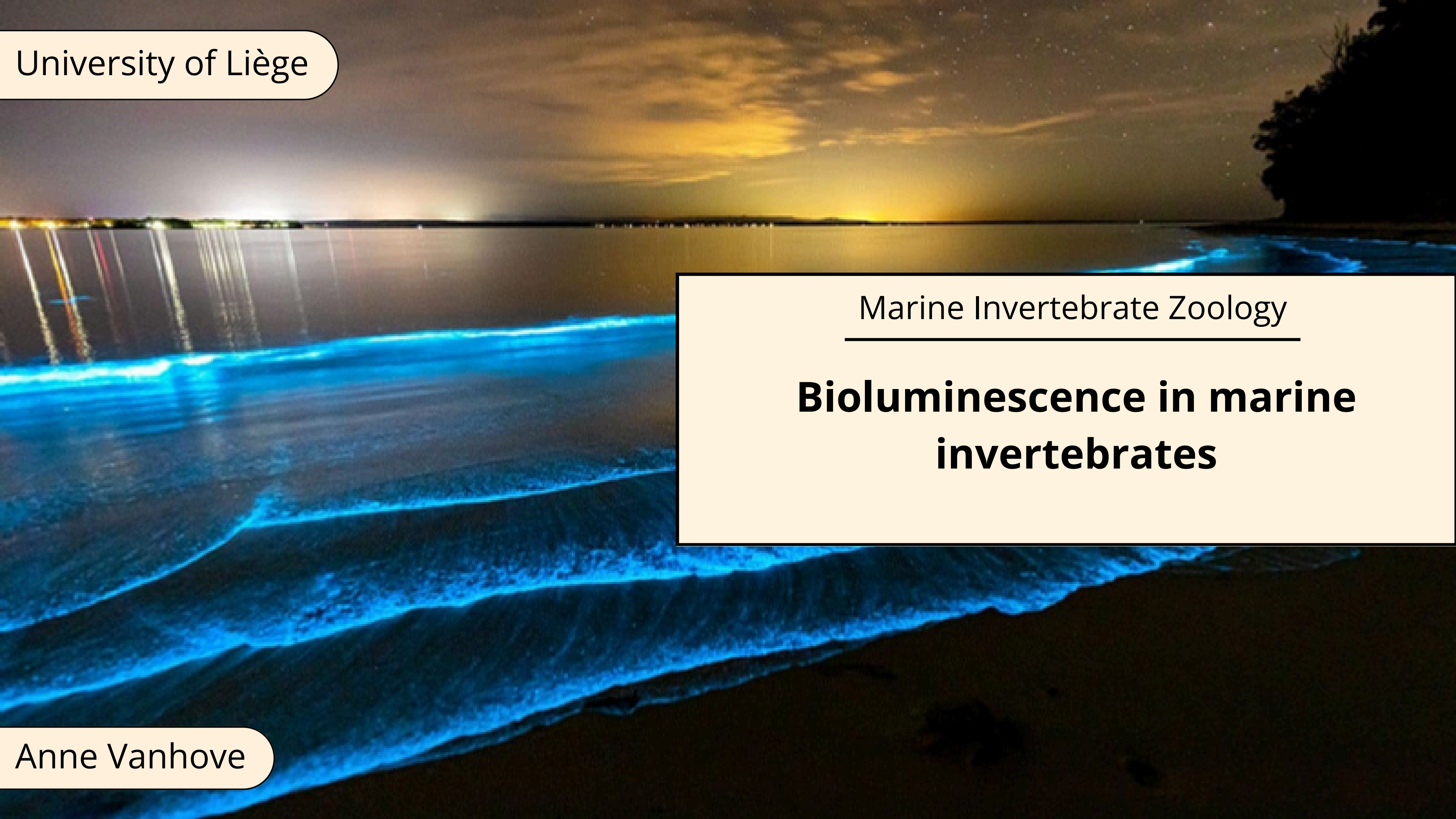




University of Liège

Marine Invertebrate Zoology

Bioluminescence in marine invertebrates

Anne Vanhove

What is bioluminescence?



Production of visible light by a living organism through a biochemical reaction (Harvey, 1952)

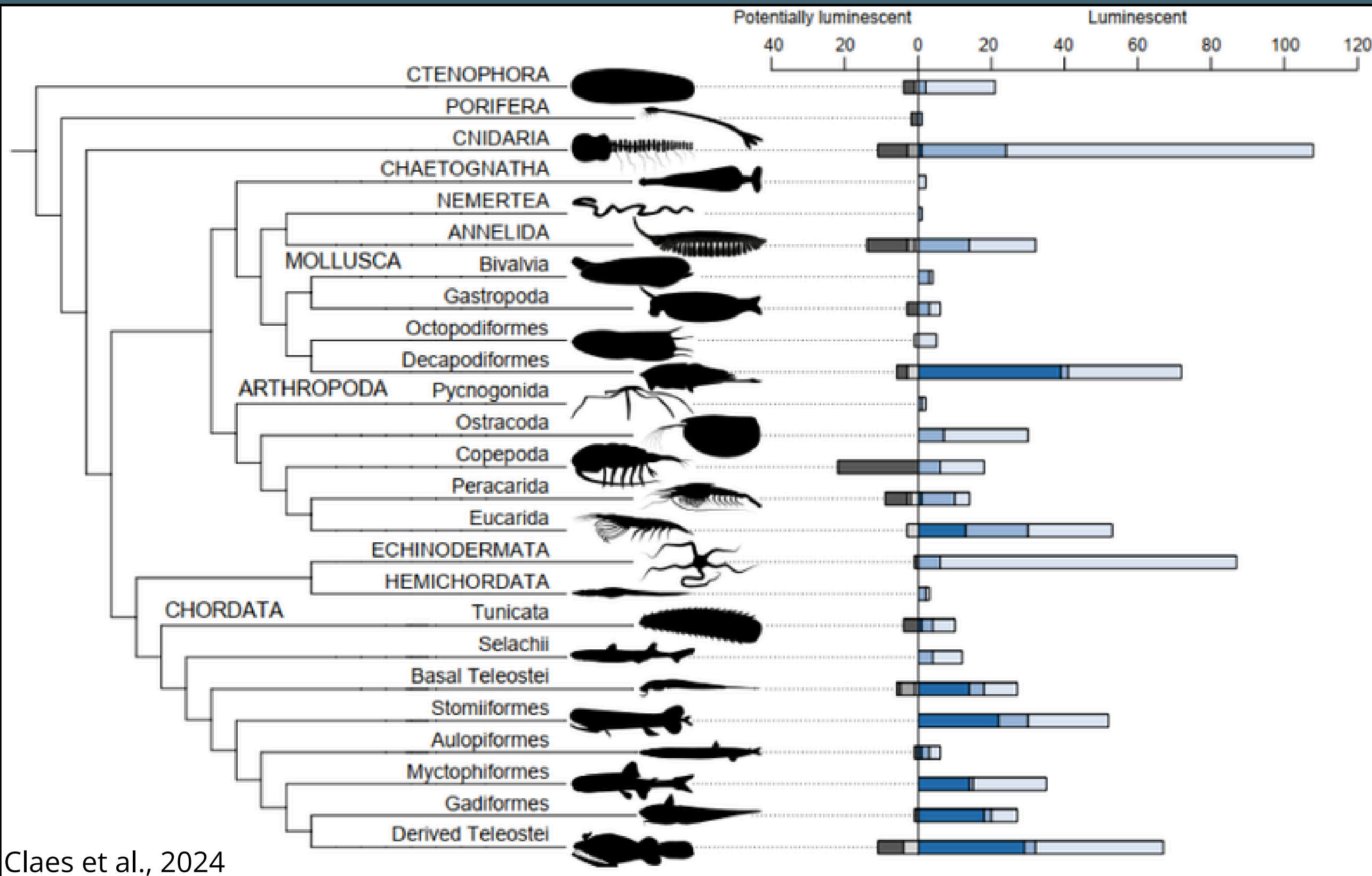
Origins and occurrences

- Appeared **94** times **independently** during evolution
- Emerged **500–600 million years ago** in marine organisms
 - **80%** are bioluminescent — an adaptation to the **environment**



Evolutionary convergence in different groups: bacteria, dinoflagellates, fish, mollusks, annelids,...

Bioluminescent organisms



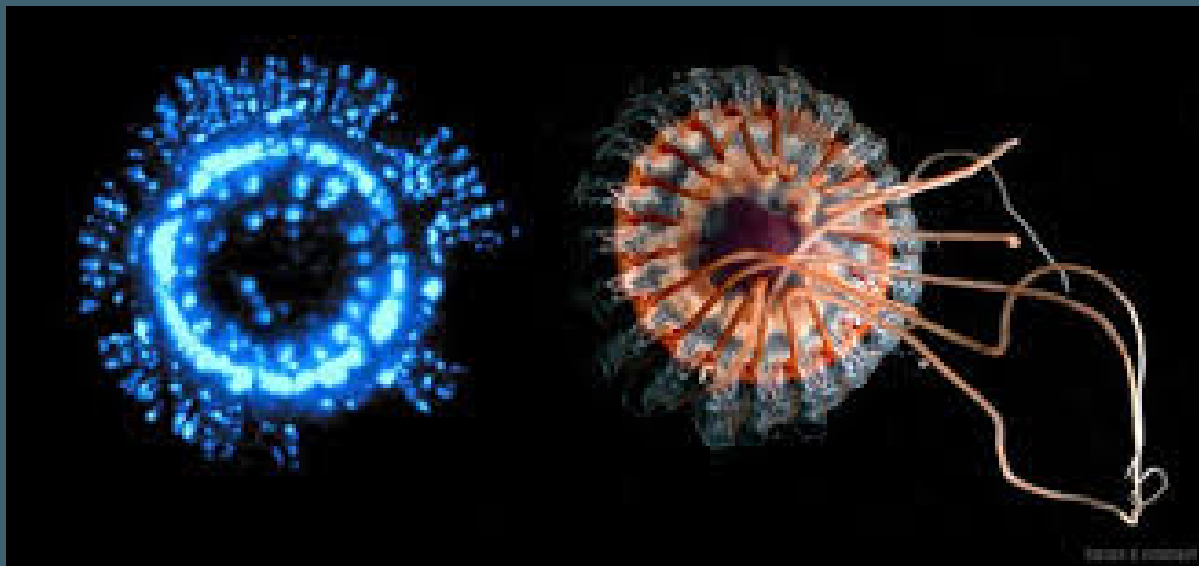
- Chordates
 - **Stomiiformes**
 - Myctophiformes
 - Some teleosts
- Arthropods
 - **Eucarids**
- Mollusks
 - **Decapodiformes**
- Annelids
- Cnidarians
- Echinoderms

What type of emission?

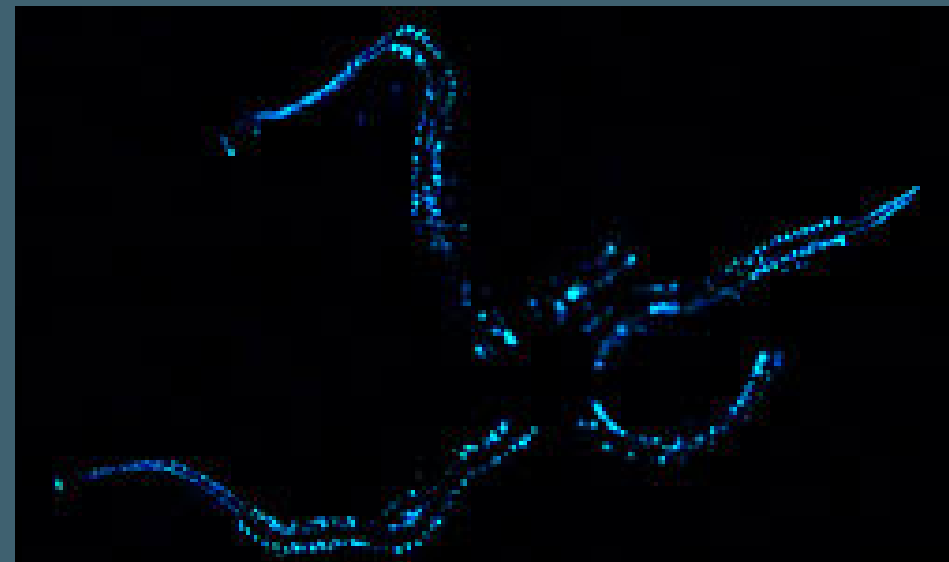
The color of bioluminescence is related to the habitat of the emitting organism:

- Violet-blue (400–500 nm) in the deep ocean
- Blue-green (460–520 nm) for more coastal species
- Green-yellow (520–580 nm) for terrestrial species

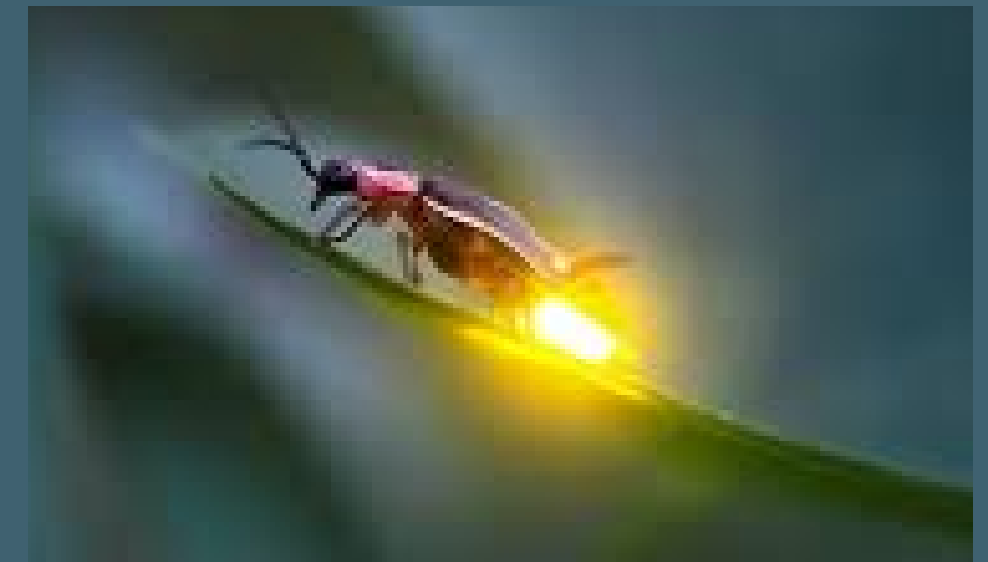
Two types of emission: glows (>3 s = attractive) and flashes (<3 s = repellent)



Atolla wyvillei
Jellyfish (*Coronatae*)
Bathypelagic zone



Amphiura filiformis
Brittle stars (*Amphiuridae*)
Epipelagic zone



Fireflies / Lampyres
(*Lampyridae*)
Terrestrial

Luciferin-luciferase

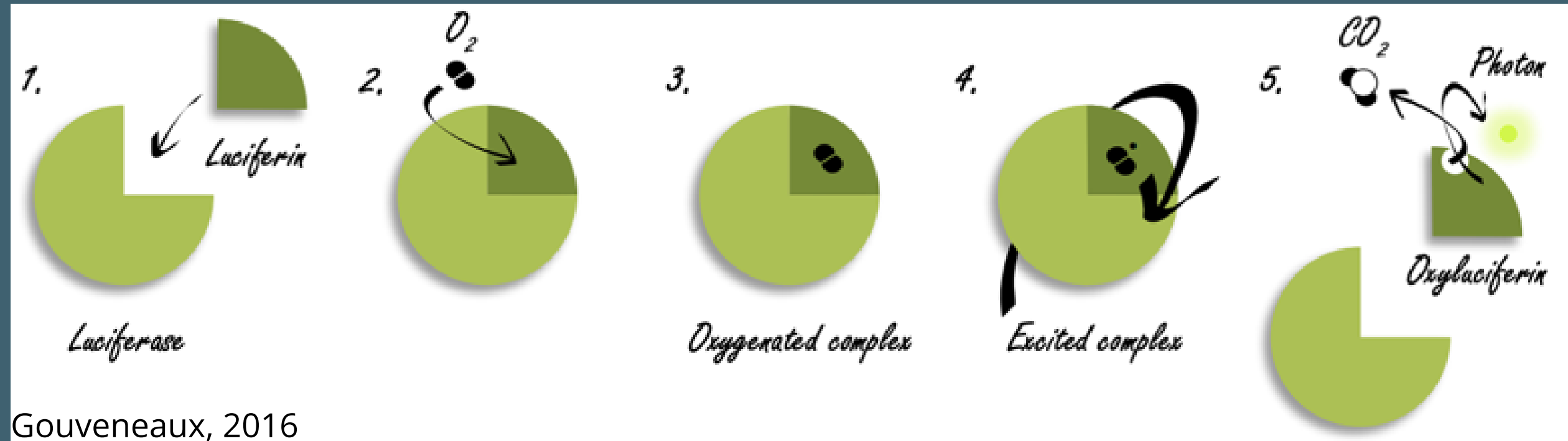
1_ **Binding** of luciferin to luciferase

2_ **O₂** binds to the luciferin-luciferase complex, forming an **oxygenated complex**

luciferin + luciferase + oxygen → oxyluciferin

3 et 4_ Formation of an **unstable** oxygenated complex that enters an **excited** state

5_ Release of a **photon** and **CO₂**

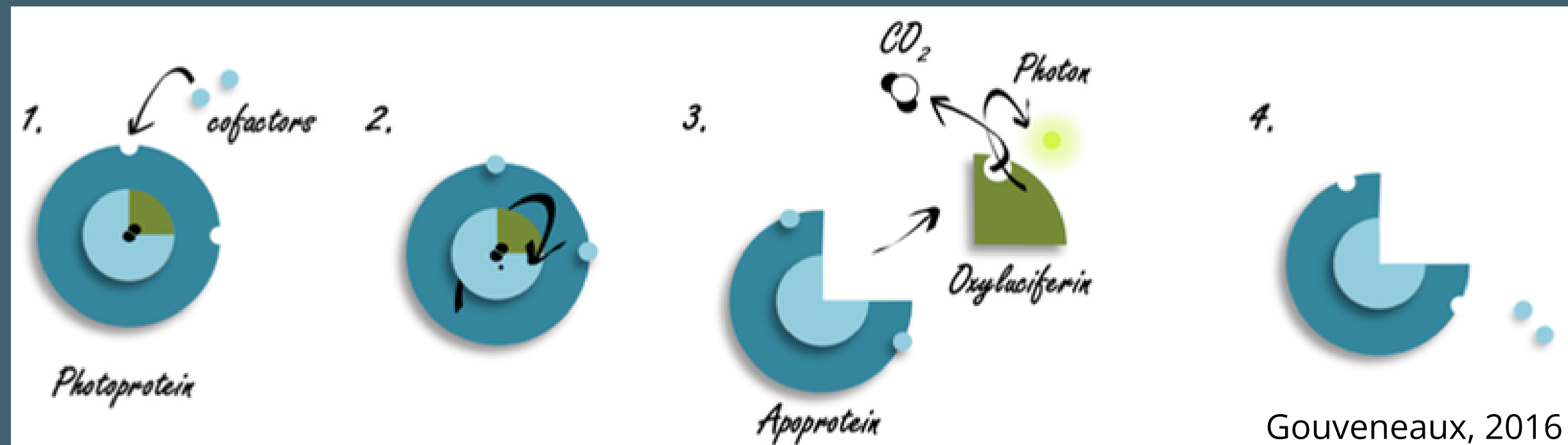


Varies among groups: bacterial luciferin, dinoflagellates, fungal luciferin, fireflies; coelenterazine in lanternfish, crustaceans, and jellyfish

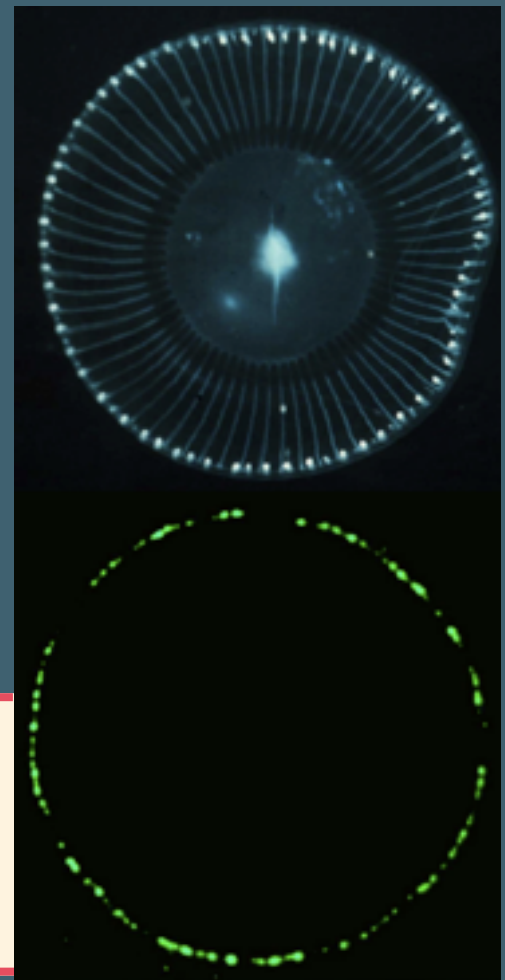
Pre-charged system

The **photoprotein** = luciferase activity + binding to luciferin and oxygen

- 1_ Activation by a **cofactor** (Ca^{2+} , Mg^{2+} , Fe^{2+})
- 2_ **Conformational change** of the photoprotein
- 3_ Substrate and oxygen molecules interact with the enzymatic part
 - > Forms an **unstable protein complex** = apoprotein
 - > Releases a **photon** and **CO_2**
- 4_ Inactive apoprotein



Aequorin was the first photoprotein, discovered in the jellyfish *Aequorea victoria* in 1962



Bioluminescence and morphological adaptations

Photocyte (cell)



- Smallest functional unit of light emission
- Located alone, in groups, or dispersed
- On superficial tissues, within or near the ectoderm
- Found in cnidarians, nemerteans, ctenophores, and echinoderms

Glandular organ



- **Organization** similar to photophores
 - BUT often larger and fewer in number
- Contains a **symbiont**
- **Control** of light emission:
 - through conditions within the glandular organ
 - by occlusion

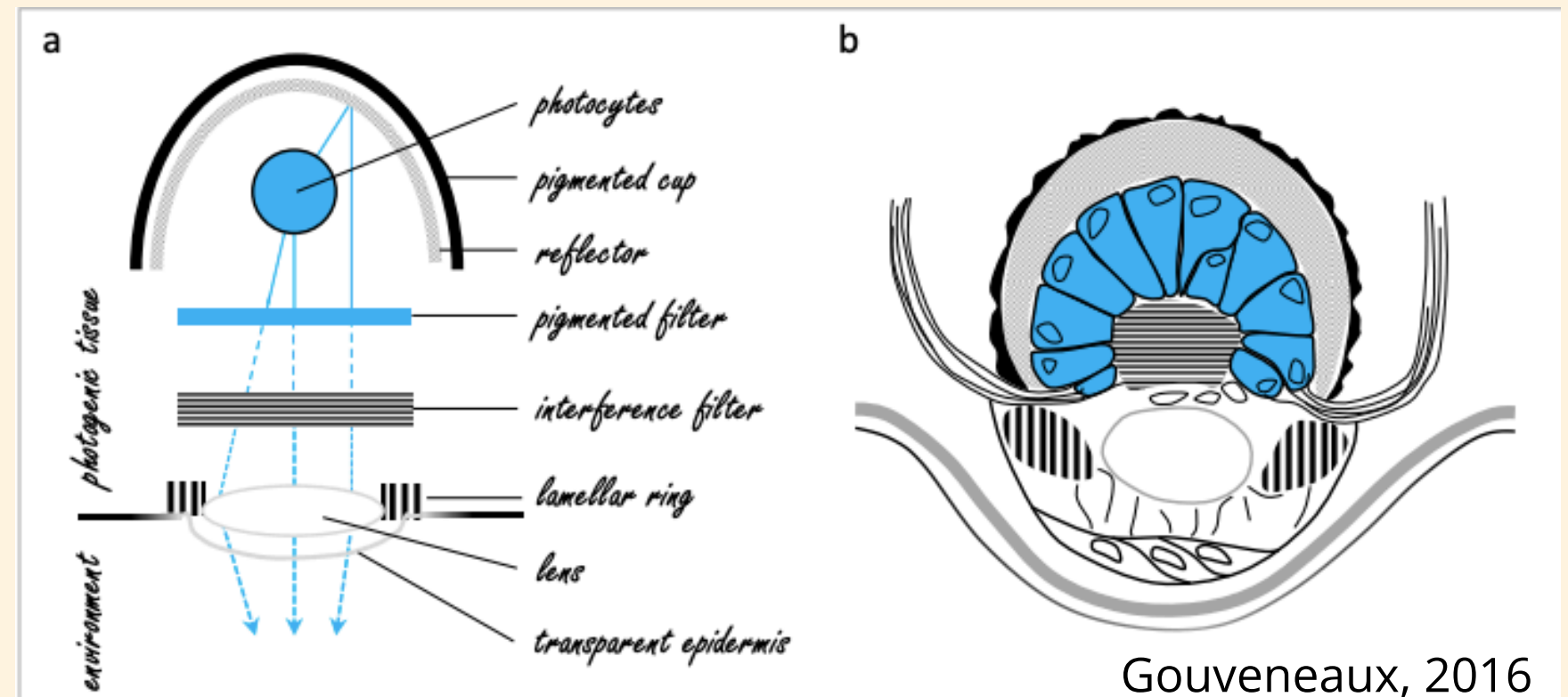
Photophore (organ)



Structure:

- Central photogenic region
 - photocytes arranged in a rosette
- Surrounded by structural elements:
 - pigmented cells, reflectors, lenses, filters

Highly developed in crustaceans, cephalopods and fish



Bioluminescence and behaviors

Predation :

Luminous lure
Attraction
Lophiiformes



Dazzle / disorient
Squid *Taningia danae*



Illuminate prey
Stomiidae
Black dragonfish

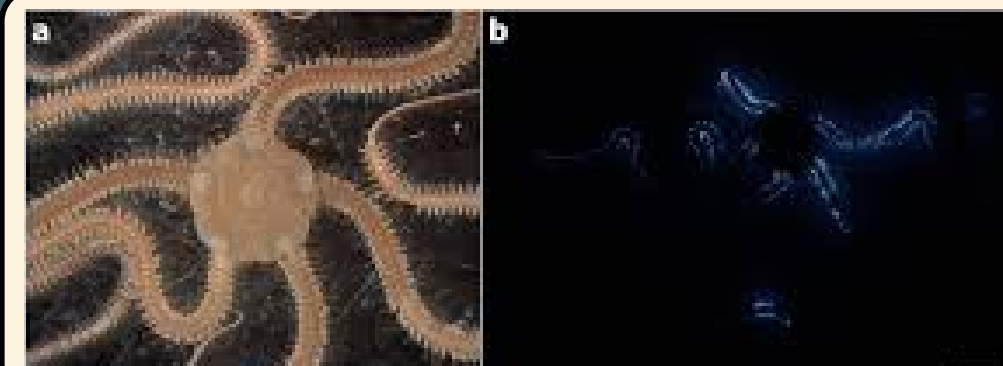


Defense :

Flash or light cloud
Crustaceans

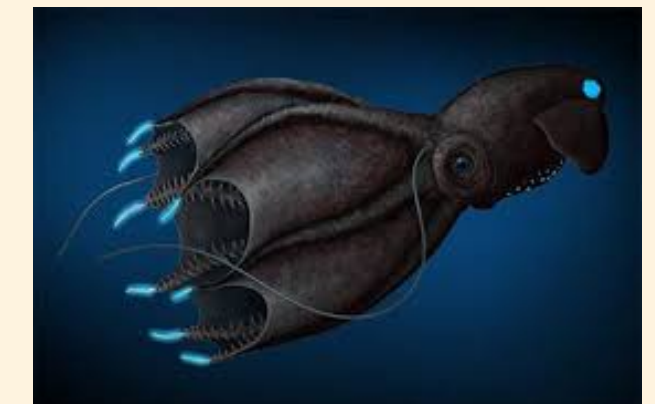


Counter-illumination
Cephalopods, fish



Sacrificial lure
Autotomy
Brittle stars, Annelids

Attract a secondary
predator





Bioluminescence and behaviors

Communication: sexual recognition and mating

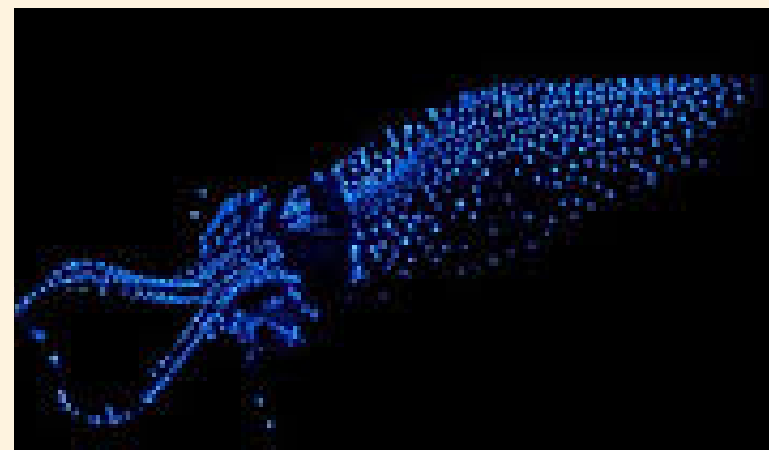
Based on observations (sometimes incorrectly)

- **Sexual dimorphism** linked to luminescence
- **Unisexual** distribution of photophores
- **Synchronized maturation**: gonads and light organs
- Light emission occurs only during the **reproductive season**



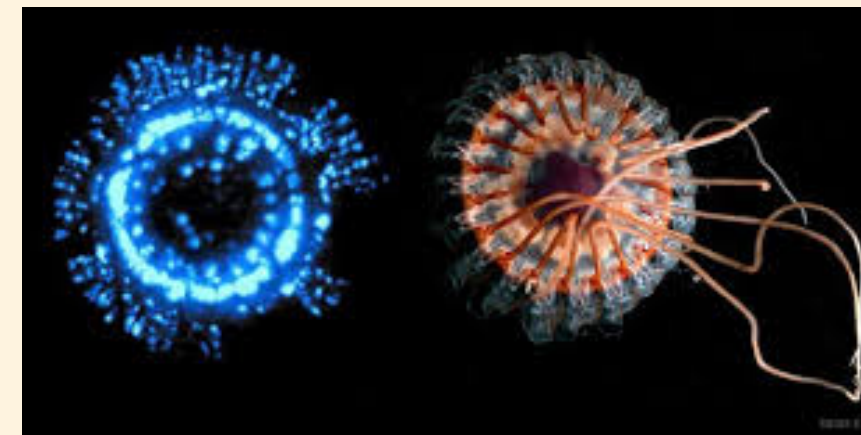
Firefly ostracods
Vargula hilgendorfii

Patterns differ by species,
♂ attract ♀



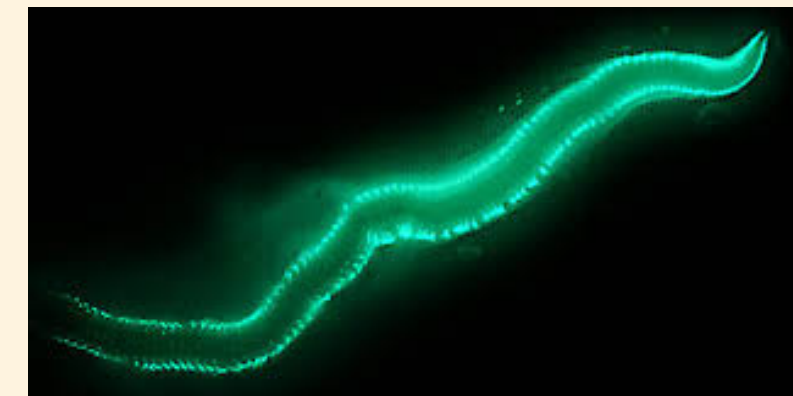
Firefly squid
Watasenia scintillans

Light pattern
→ aggregation



Red crown jellyfish
Atolla wyvillei

Aggregation
Signals a danger



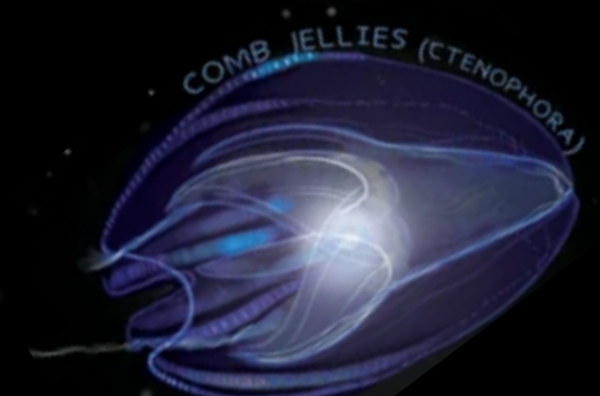
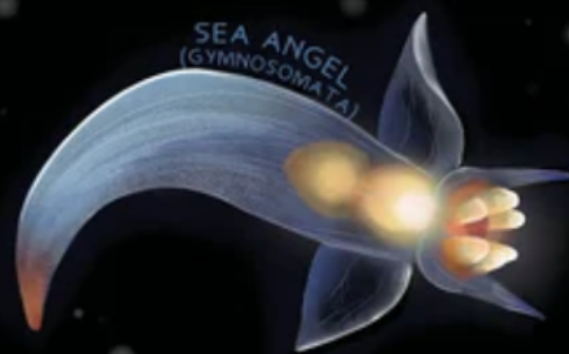
Polychaetes
Odontosyllis enopla

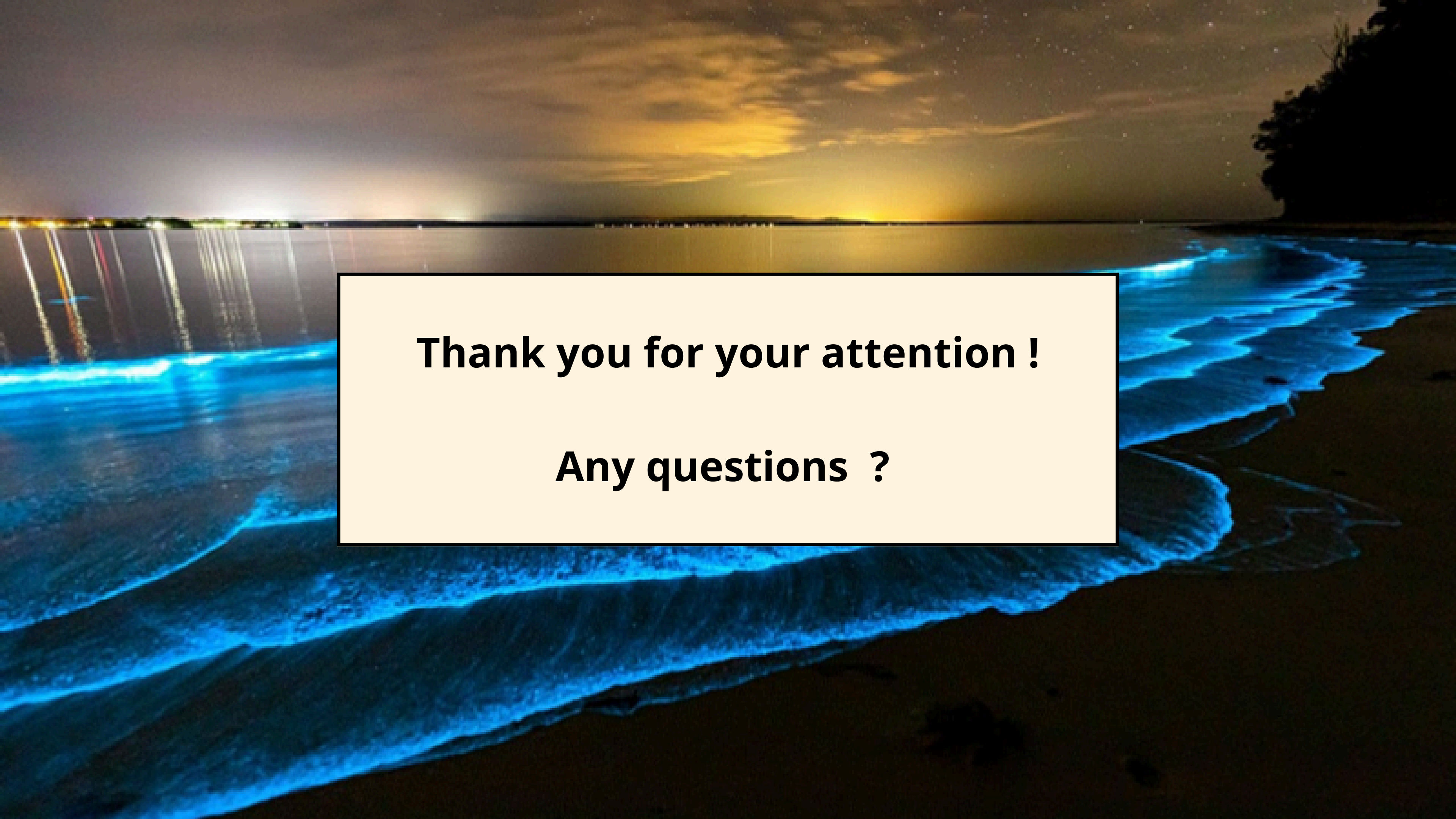
♀ secrete mucus to
attract ♂

Conclusion



- Appeared **independently** in multiple clades
 - illustrates **evolutionary convergence** influenced by phylogeny
 - Based on **biochemical mechanisms** (luciferin–luciferase) and **physiological adaptations** (photophores)
 - **Structures** serve key **behaviors**: predation, defense, and communication
 - Essential for survival in **dark** environments
- An **integrative adaptation** linking chemistry, physiology, and ecology to maximize the evolutionary success of marine invertebrates





Thank you for your attention !

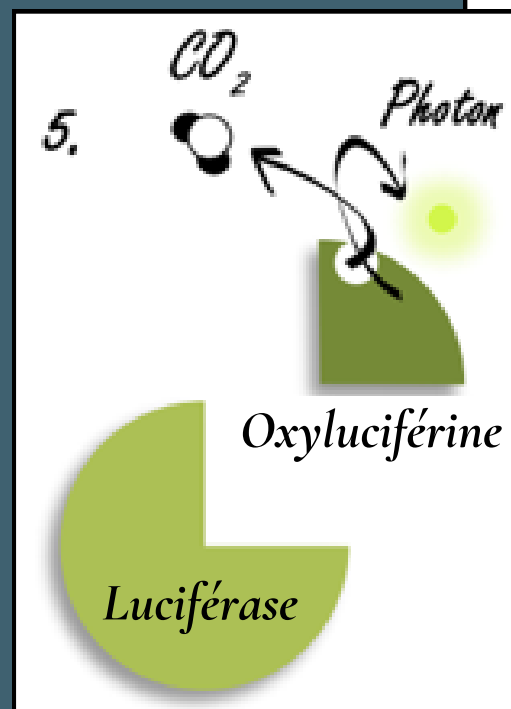
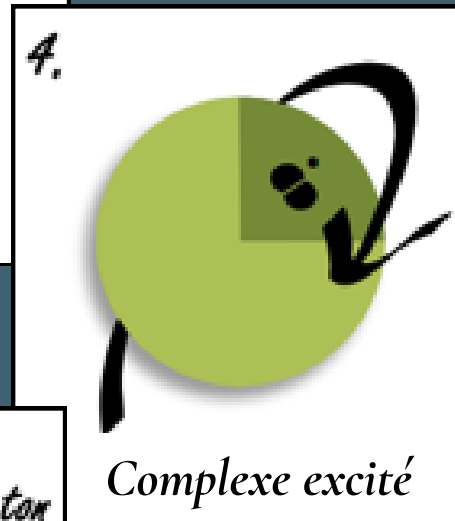
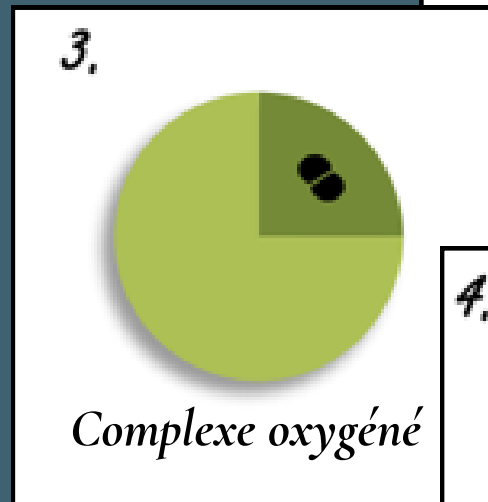
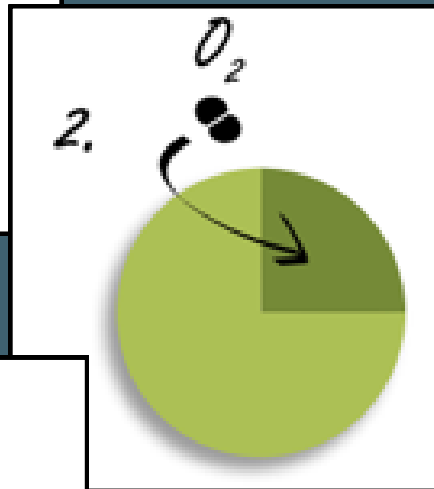
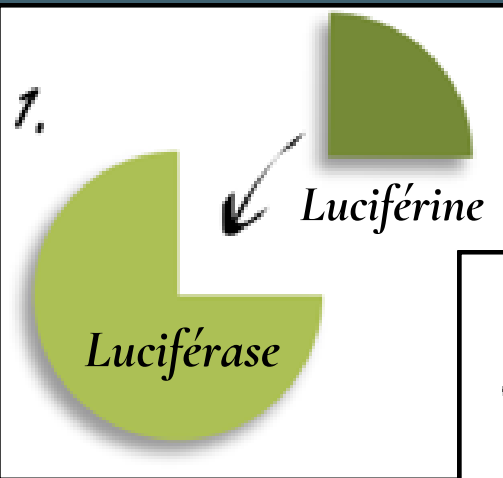
Any questions ?

References

- Collins, S. B., & Bracken-Grissom, H. D. (2024). The language of light: a review of bioluminescence in deep-sea decapod shrimps. *Biological Reviews*.
- Claes, J. M., Haddock, S. H., Coubris, C., & Mallefet, J. (2024). Systematic distribution of bioluminescence in marine animals: a species-level inventory. *Life*, 14(4), 432.
- Daunert, S., & Deo, S. K. (2006). *Photoproteins in bioanalysis*. Wiley-VCH, 240.
- Gouveneaux, A. (2016). *Bioluminescence of Tomopteridae species (Annelida): multidisciplinary approach* (Doctoral dissertation, Doctoral dissertation, Centre National de la Recherche Scientifique, Université catholique de Louvain, Belgium).
- Haddock, S. H. D., Moline, M. A., & Case, J. F. (2010). Bioluminescence in the sea. *Annual Review of Marine Science*, 2(1), 443-493.
- Hastings, J. W. (1983). Biological diversity, chemical mechanisms, and the evolutionary origins of bioluminescent systems. *Journal of molecular evolution*, 19(5), 309-321.
- Kahlke, T., & Umbers, K. D. L. (2016). Bioluminescence. *Current Biology*, 26(8), R313-R314.
- Kubodera, T., Koyama, Y., & Mori, K. (2007). Observations of wild hunting behaviour and bioluminescence of a large deep-sea, eight-armed squid, *Taningia danae*. *Biological Sciences*, 274(1613), 1029-1034.
- Martin, V. J. (2002). Photoreceptors of cnidarians. *Canadian Journal of Zoology*, 80(10), 1703-1722.
- Martini, S., & Haddock, S. H. (2017). Quantification of bioluminescence from the surface to the deep sea demonstrates its predominance as an ecological trait. *Scientific reports*, 7(1), 1-11.
- Young, R. E. (1983). Oceanic bioluminescence: an overview of general functions. *Bulletin of Marine Science*, 33(4), 829-845.
- Shimomura, O. (1981). A new type of ATP-activated bioluminescent system in the millipede *Luminodesmus sequoiae*. *FEBS Letters*, 128(2), 242-244.
- Shimomura, O. (1985). Bioluminescence in the sea: photoprotein systems. In *Symposia of the Society for Experimental Biology*, 39, 351-372).
- Shimomura, O. (2005). The discovery of aequorin and green fluorescent protein. *Journal of Microscopy (Oxford)*, 217(1), 3-15.
- Shimomura, O. (2012). *Bioluminescence: chemical principles and methods*. World Scientific.
- Widder, E. A. (2002). Bioluminescence and the pelagic visual environment. *Marine and Freshwater Behaviour and Physiology*, 35(1-2), 1-26.
- Widder, E. A. (2010). Bioluminescence in the ocean: Origins of biological, chemical, and ecological diversity. *Science (American Association for the Advancement of Science)*, 328(5979), 704-708.
- Wilson, T., & Hastings, J. W. (1998). Bioluminescence. *Annual review of cell and developmental biology*, 14(1), 197-230.

Luciférine-luciférase

Système pré-chargé



1_ **Liaison**
luciférine/luciférase

2_ Liaison **O2**
complexe oxygéné
→ **Oxyluciférine**

3_ Complexe **instable**
4_ Etat **excité**

5_ Libération
photon + CO2

Photoprotéine = activité luciférasique + liaison à la luciférine
et à l'oxygène

1_ **Liaison** co-facteur
 Ca^{2+} , Mg^{2+} , Fe^{2+}

2_ Changement de
conformation

3_ **Complexe instable**
Libération **photon + CO2**

4_ Apoprotéine **inactive**

